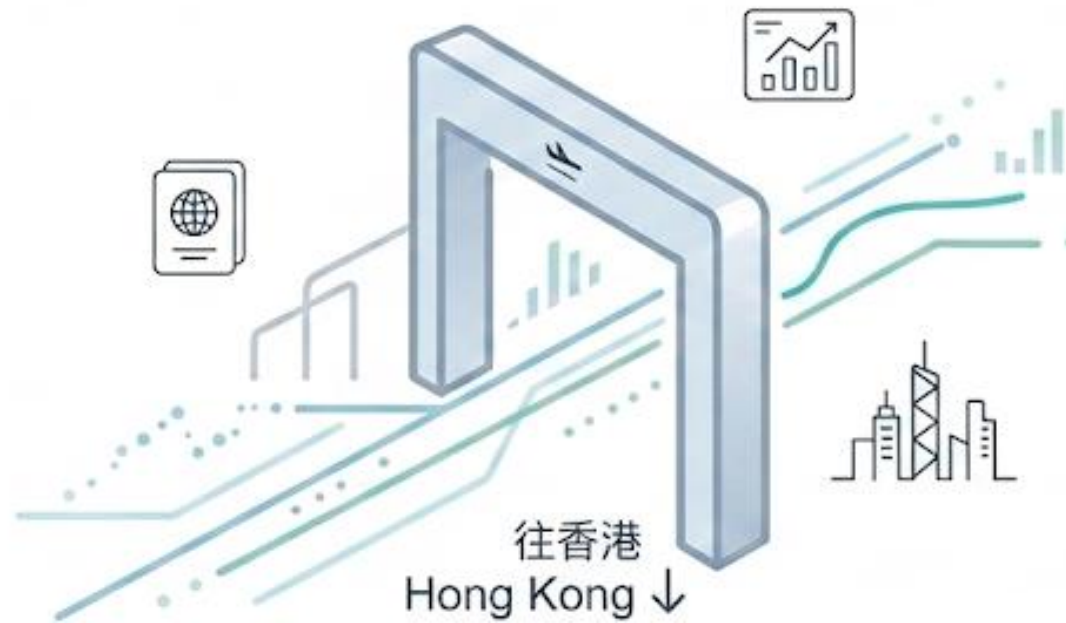


Silence and Revival: Data Visualization on Passenger Flow at HK Ports



**Group
Members**

Data12

HUANG Chudan, XIONG Yifan,
LIANG Genming, Tong Shuwei,
CHEN Hongyu

CONTENT

1

Background

2

**Data
Structure**

3

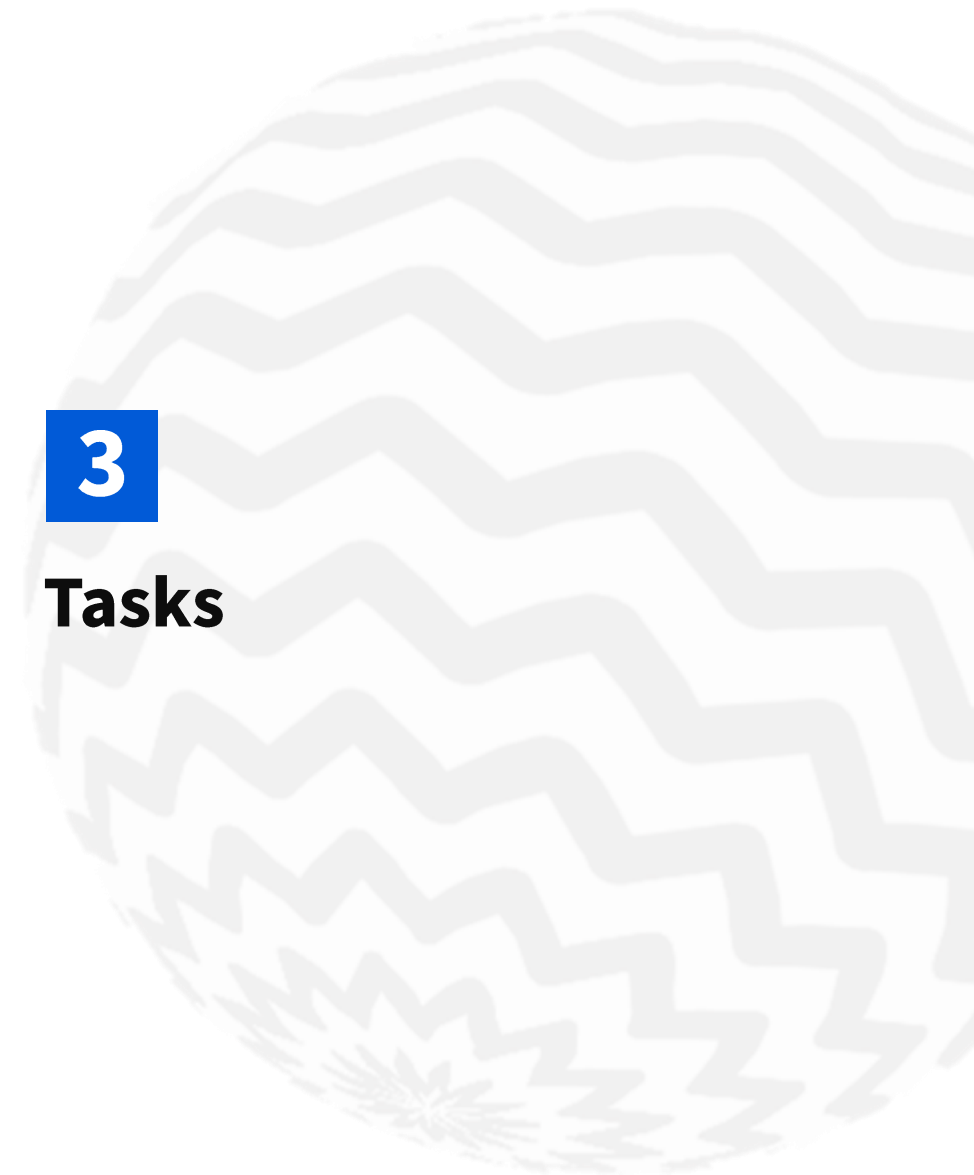
Tasks

4

Design&Implement

5

Demo





1 Background



Time span

2019 – 2026



Focus

Hong Kong
port passenger flow



Trend

decline → recovery



Goal

Reveal the pulse of
cross-border
mobility



2019 and
before..

Mid 2020 - 2023
Silence Period

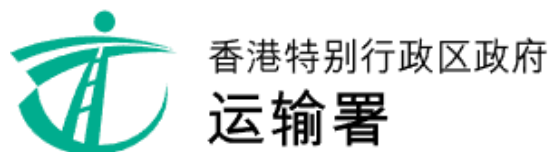
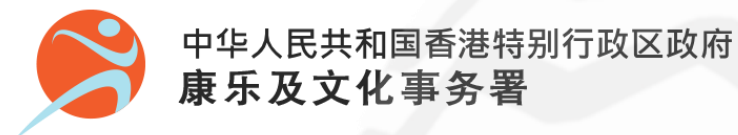
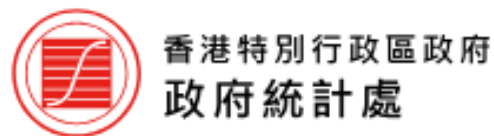
2023 and after
Revival Period

Visualization uses
dynamic charts and
maps to tell the story
of silence and revival.



2 Data Structure

Data Source



1. Daily Passenger Arrivals and Departures Statistics
2. Vehicle Cross-Border Statistics by Port
3. HKD-RMB Historical Exchange Rates
4. Visitor Flow Data for Disneyland / Ocean Park / Museums
5. Monthly Hotel Room Occupancy Rate
6. Monthly Restaurant Receipts
7. Composite Consumer Price Index (CCPI)





2 Data Stucture

Overview

65k items
2.5k days
16 HK ports
1B passengers
50M vehicles

日期	管制站	入境 / 出境	香港居民	内地访客	其他访客	总计			
14-11-2024	启德邮轮码头	出境	54	0	51	105			
14-11-2024	港澳客轮码头	入境	2021	545	1242	3808			
14-11-2024	港澳客轮码头	出境	2170	837	1506	4513			
15-11-2024	机场	入境	29881	9327	22938	62146			
15-11-2024	机场	出境	37354	7842	21638	66834			
15-11-2024	高铁西九龙	入境	15736	17216	2070	35022			
15-11-2024	高铁西九龙	出境	26337	13679	1900	41916			
15-11-2024	罗湖	入境	66156	12500	1056	79712			
15-11-2024	罗湖	出境	82266	14449	1071	97786			
15-11-2024	落马洲支线	入境	51759	13952	538	66249			
15-11-2024	落马洲支线	出境	67825	13020	516	81361			
15-11-2024	香园围	入境	25771	6226	162	32159			
15-11-2024	香园围	出境	30078	3738	261	34077			
15-11-2024	港珠澳大桥	入境	18667	5294	5115	29076			
15-11-2024	港珠澳大桥	出境	30596	9170	4010	43776			
25-11-2024	港珠澳大桥	出境	2353	651	13873				
25-11-2024	港珠澳大桥	出境	3296	495	15228				
LMC	201301	IB	25	13203	371	27	2265		
LMC	201301	OB	1	144363	320	9	2459		
LMC	201301	OB	23	21931	0	0	0		
LMC	201301	OB	25	13349					
LMC	201301	IB	1	154734					
LMC	201301	OB	36	209221					
LMC	201301	IB	日期	收盘	开盘	高	低	交易量	涨跌幅
LMC	201301	IB	2026/4/1	0.8782	0.88	0.8804	0.8767		-0.20%
LMC	201302	IB	2026/3/1	0.88	0.8767	0.8879	0.8765		0.38%
LMC	201302	IB	2026/2/1	0.8767	0.8898	0.8906	0.8734		-1.46%
LMC	201302	IB	2026/1/1	0.8897	0.8986	0.8999	0.8895		-0.99%
LMC	201302	OB	2025/12/1	0.8986	0.9087	0.9092	0.8975		-1.11%
LMC	201302	OB	2025/11/1	0.9087	0.9159	0.9174	0.9086		-0.79%
			2025/10/1	0.9159	0.9148	0.9187	0.9131		0.12%
			2025/9/1	0.9148	0.9145	0.9174	0.9132		0.03%
			2025/8/1	0.9145	0.9173	0.9217	0.9131		-0.31%
年月	指数类别	组别类别	基数年	指数	权数		0.9107		0.52%
201301	CCPI	12	Index : October 2014	96.6	7.98		0.9118		-0.61%
201301	CCPI	99	Index : October 2014	91.9	100		0.9149		-2.07%
201301	CCPI	1	Index : October 2014	94.7	0.4		0.9327		0.51%
201301	CCPI	2	Index : October 2014	111.1	0.55		0.9286		-0.38%
201301	CCPI	3	Index : October 2014	93.7	1.76		0.9302		0.62%
201301	CCPI	4	Index : October 2014	95	1.18				
201301	CCPI	5	Index : October 2014	100.3	0.01				
201301	CCPI	6	Index : October 2014	95.2	0.41				
201301	CCPI	7	Index : October 2014	93.2	0.43				
201301	CCPI	8	Index : October 2014	93.8	1.46				
201301	CCPI	9	Index : October 2014	95.5	0.06				
201301	CCPI	10	Index : October 2014	85.2	0.24				
201301	CCPI	11	Index : October 2014	103	1.48				
201302	CCPI	12	Index : October 2014	97.1	7.98				
201302	CCPI	99	Index : October 2014	92.8	100				

Data Composition

- Dimension Fields (6 items)
- Year
 - Month
 - Port
 - Port Type
 - Traffic Data Available
 - Direction of Travel (Inbound/Outbound)

- Passenger Data (4 items)
- Hong Kong Residents
 - Mainland Visitors
 - Other Visitors
 - Total Passengers

- Growth Rates (6 items)
- Month-on-Month Passenger Growth
 - Year-on-Year Passenger Growth
 - MoM Growth of Mainland Visitors
 - YoY Growth of Mainland Visitors
 - MoM Traffic Growth
 - YoY Traffic Growth

- Vehicle Traffic Data (4 items)
- Private Cars
 - Tour Coaches
 - Shuttle Buses
 - Freight Vehicles



3 Tasks

Task 1: Rational Visualization of Passenger Flow at Ports

- 1. Visualize passenger volume**
- 2. Show flow distribution, peak hours, holiday surges, and recovery trends**
- 3. Decide chart types.**

Task 2: Analyze Socio-economic Impacts of Passenger Flow

- 1. Explore correlations between passenger flow and Hong Kong 's economy**
- 2. Key dimensions: Consumption & Transportation & Attraction**
- 3. Output: Correlation charts, scatter plots, heatmaps, trend overlays**





4 Desgin & Implement

日期	管制站	入境 / 出境	香港居民	内地访客	其他访客	总计
14-11-2024	启德邮轮码头	出境	54	0	51	105
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15-11-2024	落马洲支线	出境	67825	13020	516	81361
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15-11-2024	落马洲	入境	10869	2353	651	13873
15-11-2024	落马洲	出境	11437	3296	495	15228
15-11-2024	文锦渡	入境	1867	371	27	2265
15-11-2024	文锦渡	出境	2130	320	9	2459

关卡	年月	行驶方向	车辆种类	车流量
LMC	201301	IB	25	13203
LMC	201301	OB	1	144363
LMC	201301	OB	23	21931
LMC	201301	OB	25	13349
LMC	201301	IB	1	154734
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LMC	201301	IB	36	238566
LMC	201301	IB	23	21829

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2026/4/1	0.8782	0.88	0.8804	0.8767		-0.20%
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2025/9/1	0.9148	0.9145	0.9174	0.9132		0.03%
2025/8/1	0.9145	0.9173	0.9217	0.9131		-0.31%
2025/7/1	0.9173	0.9126	0.9173	0.9107		0.52%
2025/6/1	0.9126	0.9182	0.9183	0.9118		-0.61%
2025/5/1	0.9182	0.9376	0.9383	0.9149		-2.07%

年月	指数类别	组别类别	基数年	指数	权数
201301	CCPI	12 Index : October 2014		96.6	7.98
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Single Dimension Statistics (Single Metric)



Multi-dimensional Analysis (20 Metrics)



Spatio-temporal Cross Analysis (16 × 58 × 2)

DATA
CLEANING

Deduplication

Validation

Standardization

Outlier Handling

DATA
AGGREGATION

Time Dimension
Aggregation

Vehicle Flow
Data Integration

FEATURE
ENGINEERING

Dimension Field
Extraction

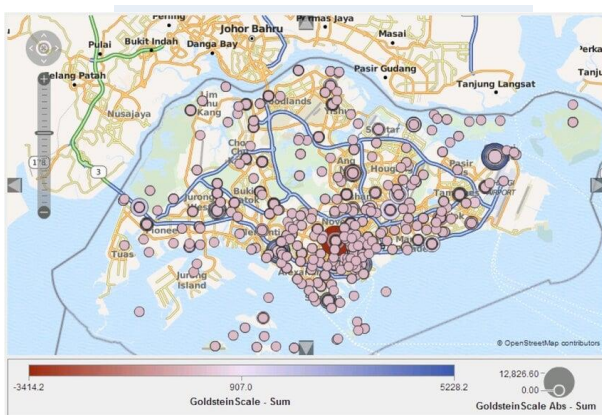
Growth Rate
Calculation

Fix Infinite Values



4 Design & Implement

TYPE 1: BUBBLE MAP



- Time-series geographic distribution map
- Visualization of passenger flow trends by port
- Realtime flow map animation



MAP VISUALS & TIME SERIES

Passenger distribution across ports. Dynamic charts for inbound/outbound animation

TYPE 2: MULTI-LAYER PROPORTION



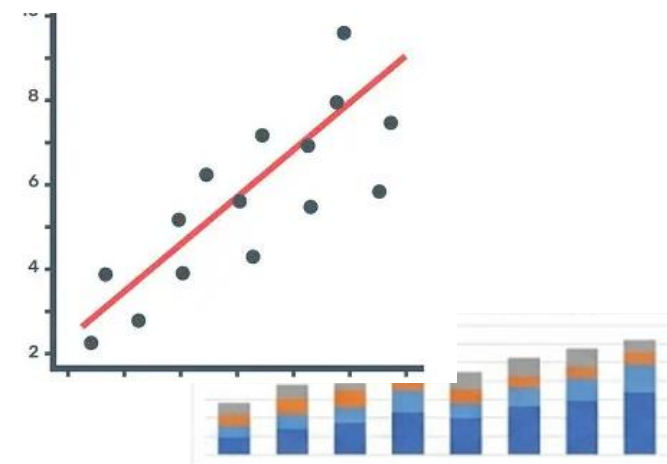
- Hierarchical port share display
- Multi-level category comparison
- Flow composition structure analysis



DYNAMIC HIERARCHY

Multi-level data share
Drill-down composition
Dynamic portion view

TYPE 3: CORRELATION & TRENDS



- Correlation metrics
- Multi-variable analysis
- Time-series line and area charts
- Period comparisons



INSIGHTS

Charts for analyzing historical trends, seasonal patterns, and direct comparisons.



5 Demo

- Overview
- Northern Land Ports
- Western Ports
- Central Urban Ports

Airport

HK-Zhuhai-Macao Bridge

XRL West Kowloon

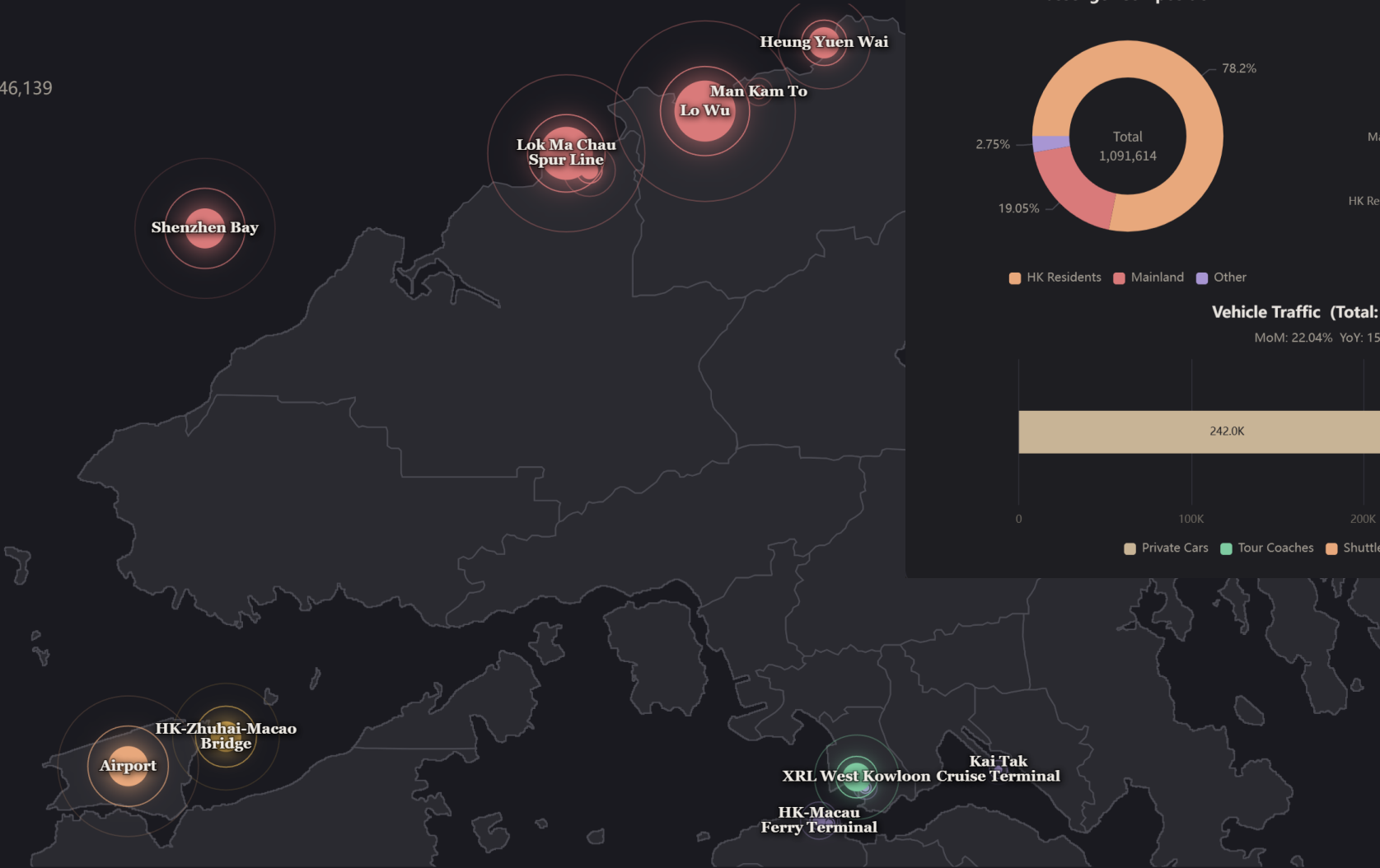
Land Ports

Sea Ports

Ripple size = Total Passengers

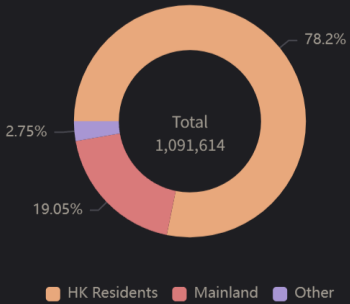
2025-03

Monthly Total Passengers: 27,046,139



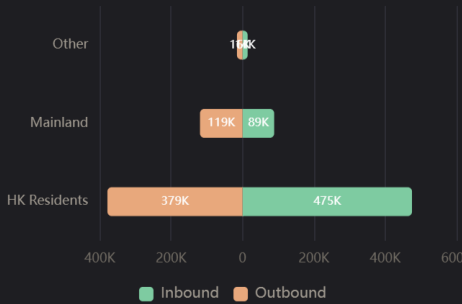
Lok Ma Chau 2025-03

Passenger Composition



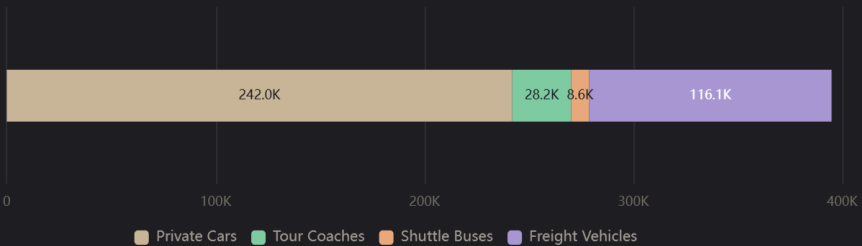
Inbound vs Outbound

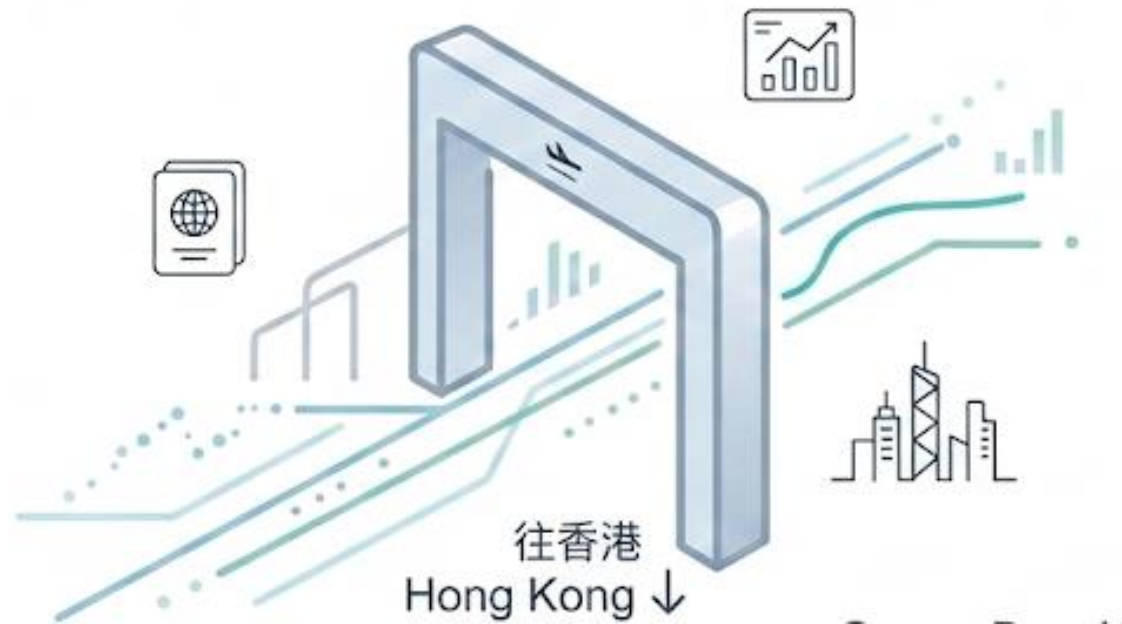
Pax MoM: 11.65% YoY: 15.2%



Vehicle Traffic (Total: 394,928)

MoM: 22.04% YoY: 15.84%





Thank you

**Group
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